How AI DeepGuard Detection Works

The complete system can be explained in the following sequence:

1. User Login / Authentication

- The user first logs into the system.
- The Flask backend verifies the user's credentials.
- Only authenticated users can access the detection functionality.
- Admin users can access additional administrative features.

2. Upload Image

- The user uploads an image through the web interface.
- The system accepts the image for deepfake analysis.
- Basic validation is performed before processing.

3. Image Preprocessing

- The uploaded image is prepared for AI analysis.
- Image dimensions are standardized.
- The image is normalized as required by the detection model.
- Preprocessing helps provide consistent input to the model.

4. Face Detection

- The system identifies the face present in the uploaded image.
- The relevant facial region is extracted for further analysis.
- This focuses the deepfake detector on the most important region of the image.

5. Feature Extraction

- The processed face is converted into features that can be analyzed by the deep learning model.
- The model looks for visual patterns and inconsistencies associated with manipulated facial content.

6. Deep Learning Model

- The extracted facial information is passed to the deep learning model.
- The model analyzes the learned characteristics of genuine and manipulated images.
- It produces a classification result.

7. Real / Fake Prediction

The system generates the primary result:

REAL → The image is predicted to be genuine.

FAKE → The image is predicted to be manipulated/deepfake.

The prediction is based on the model's learned patterns.

8. Confidence Score

- Along with the prediction, the system generates a **confidence score**.
- The score indicates how strongly the model supports its prediction.
- For example, the interface could show:

Prediction: FAKE

Confidence: 94%

The actual confidence value depends on the trained model and input image.

9. Deepfake Analysis

- The system provides additional analysis of the prediction.
- It helps the user understand the detection result rather than showing only "Real" or "Fake."

10. Grad-CAM / Heatmap

- **Grad-CAM** is used as an Explainable AI (XAI) technique.
- It generates a heatmap over the facial image.
- Important regions contributing to the model's prediction are highlighted.
- This provides a visual explanation of **where the model is focusing**.

Conceptually:

Original Image

↓

Face Detection

↓

Deep Learning Model

↓

Prediction

↓

Grad-CAM

↓

Heatmap

11. Result Dashboard

The final result is displayed on the dashboard.

It can contain:

- Uploaded image
- Detected face
- Real/Fake result
- Confidence score
- Deepfake analysis
- Grad-CAM / Heatmap
- Detection date/time

12. Store Detection Result

- The detection information is stored in the database.
- This allows the system to maintain records of previous analyses.
- User-specific detection records can be retrieved later.

13. Detection History

The user can view previous detections.

For example:

Detection History

Image	Prediction	Confidence	Date
Image 01	FAKE	94%	...
Image 02	REAL	87%	...
Image 03	FAKE	91%	...

14. Report Generation

- The system can generate a report for a completed detection.
- The report can contain the image, prediction, confidence score, analysis and heatmap.
- This makes the result easier to save or present.

15. Admin Dashboard

The administrator has additional system-management functionality.

The Admin Dashboard can be used to manage:

- Users
- Authentication-related records
- Detection records
- Database information
- Overall system activity

Complete Working Flow

For your **viva**, explain it like this:

USER

↓

LOGIN / AUTHENTICATION

↓

UPLOAD IMAGE

↓

IMAGE PREPROCESSING

↓

FACE DETECTION

↓

FEATURE EXTRACTION

↓

DEEP LEARNING MODEL

↓

REAL / FAKE PREDICTION

↓

CONFIDENCE SCORE

↓

DEEPFAKE ANALYSIS

↓

GRAD-CAM / HEATMAP

↓

RESULT DASHBOARD

↓

DATABASE

↓

DETECTION HISTORY

↓

REPORT GENERATION

Technology Role

Technology	Role
HTML	Web page structure
CSS	User interface styling
JavaScript	Frontend interaction and dynamic operations
Flask	Backend, routing, authentication and AI integration
Python	AI/model processing
Deep Learning	Real/Fake classification
Computer Vision	Face and image processing
Grad-CAM	Explainable AI / heatmap generation
Database	Users and detection-history storage

Research papers

1. LAA-Net — 2024

Paper: *LAA-Net: Localized Artifact Attention Network for Quality-Agnostic and Generalizable Deepfake Detection*
Author: D. Nguyen et al.
Year: 2024

What is the paper about?

- This paper focuses on detecting deepfake images by identifying **localized artifacts** created during face manipulation.
- Instead of treating the entire image equally, the approach focuses attention on regions that are more likely to contain manipulation artifacts.
- It aims to make deepfake detection more **generalizable**, meaning the detector should work better even when it encounters manipulation techniques or image qualities different from those seen during training.
- The approach uses **localized artifact attention** and multi-task learning to improve detection.

Why is it relevant to our project?

Our project also needs to identify suspicious regions of a face. The idea of focusing on localized artifacts supports our **Grad-CAM / Heatmap** feature.

Viva answer:

“LAA-Net focuses on localized artifacts in manipulated facial images. Its main contribution is improving deepfake detection by directing attention toward artifact-prone regions and improving generalization. This is relevant to our project because we also want to visually explain which regions contribute to a deepfake prediction.”

2. SIDA — 2025

Paper: *SIDA: Social Media Image Deepfake Detection, Localization and Explanation with Large Multimodal Model*
Authors: Z. Huang et al.
Year: 2025

What is the paper about?

SIDA focuses on **three important tasks**:

1. **Detection** — determine whether an image is manipulated.
2. **Localization** — identify the manipulated regions.
3. **Explanation** — provide an explanation for the detection result.

The paper is particularly focused on deepfake images encountered in **social-media environments**.

It uses a **Large Multimodal Model** and includes image detection and localization capabilities.

Why is it relevant to our project?

This paper is very closely related to our proposed system because AI DeepGuard also combines:

Detection + Analysis + Visual Explanation

Our project uses **Grad-CAM / Heatmap** to provide a visual explanation of the model's prediction.

Viva answer:

“SIDA extends deepfake detection beyond simply classifying an image as real or fake. It also focuses on localization and explanation. This inspired the explainability aspect of our project, where AI DeepGuard provides a prediction together with analysis and a Grad-CAM heatmap.”

3. Hybrid Bag-of-Visual-Words + Multi-CNN — 2026

Paper: *Deepfake Face Detection Using Hybrid Bag-of-Visual-Words and Multi-CNN Feature Fusion*
Authors: M. Alrahhal, F. Alqahtani, R. Latip, M. AlShabi, and W. M. Abd-Elhafiez
Year: 2026

What is the paper about?

- This paper investigates **face deepfake detection** using a hybrid feature approach.
- It combines traditional visual/forensic feature descriptors with features extracted by multiple CNN architectures.
- The basic idea is to combine different types of features instead of relying on only one representation.
- The paper evaluates the approach using multiple benchmark datasets.

Why is it relevant to our project?

It demonstrates the usefulness of combining **forensic visual information and deep-learning features** for face deepfake detection.

Our project similarly uses AI-based facial analysis as the core of the detection pipeline.

Viva answer:

“The 2026 paper proposes a hybrid approach that combines handcrafted visual features with features from multiple CNN models for face deepfake detection. The main idea is feature fusion, where different types of information are combined to improve detection. This is relevant to our project because our system also performs facial analysis using deep-learning-based detection.”

What these papers collectively tell us

You can explain the evolution like this:

2024

LAA-Net

↓

Focus on LOCALIZED ARTIFACTS

↓

2025

SIDA

↓

DETECTION + LOCALIZATION + EXPLANATION

↓

2026

Hybrid BoVW + Multi-CNN

↓

FORENSIC FEATURES + DEEP CNN FEATURES

And then connect them to **AI DeepGuard**:

Research Findings

↓

Localized Artifacts

+

Detection

+

Explanation

+

Deep Feature Analysis

↓

AI DEEPGUARD DETECTION

↓

Real / Fake

+

Confidence Score

+

Deepfake Analysis

+

Grad-CAM / Heatmap

+

History / Reports / Dashboard